

Exploratory Data Analysis (EDA)



918
Total Records



11
Features



0
Missing Values

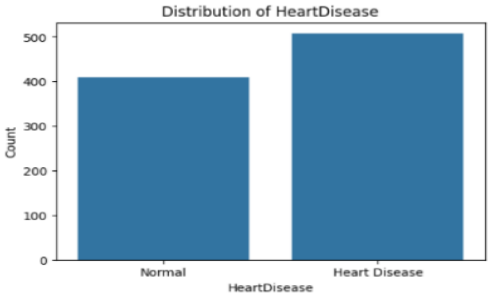


0
Duplicates

DATA QUALITY & BASELINE

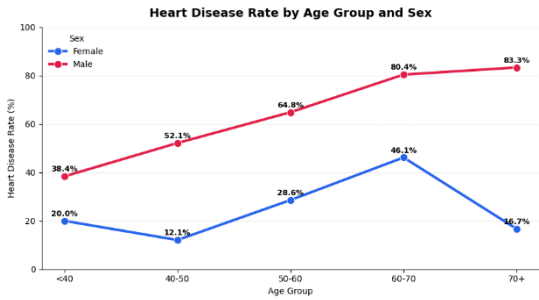
	Age	RestingBP	Cholesterol	MaxHR	Oldpeak
count	918.000000	918.000000	918.000000	918.000000	918.000000
mean	53.510893	132.396514	198.799564	136.809368	0.887364
std	9.432617	18.514154	109.384145	25.460334	1.066570
min	28.000000	0.000000	0.000000	60.000000	-2.600000
25%	47.000000	120.000000	173.250000	120.000000	0.000000
50%	54.000000	130.000000	223.000000	138.000000	0.600000
75%	60.000000	140.000000	267.000000	156.000000	1.500000
max	77.000000	200.000000	603.000000	202.000000	6.200000

Invalid zero values were found in RestingBP (1 row) and Cholesterol (172 rows); RestingBP was removed, while Cholesterol was imputed.

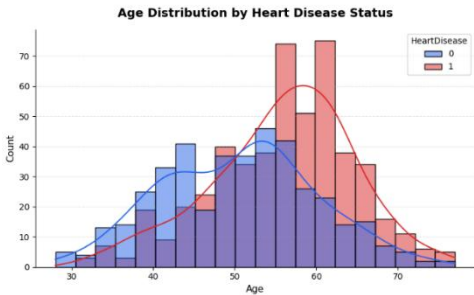


Dataser target balanced with heart disease higher than normal

DEMOGRAPHIC RISK PROFILING

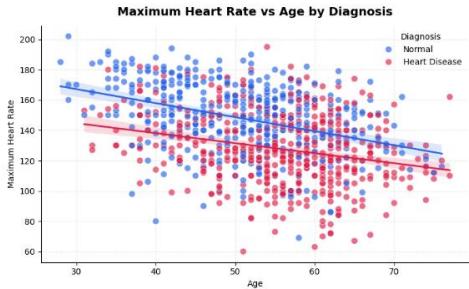


Female patients have consistently higher heart disease prevalence across all age group. Risk increases exponentially after age 50.

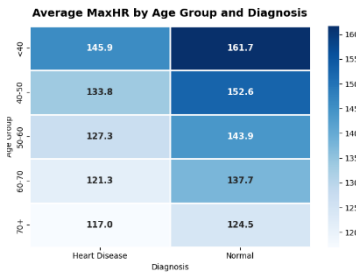


Heart disease prevalence peaks in the 55-65 age range, indicating age is a key risk factor

INDICATORS



MaxHR generally declines with age in both groups, but heat disease patients have lower MaxHR at nearly all ages.



MaxHR is consistenlu lower for heart disease patients in every age group, with the largest gap in the <50 age range

EXECUTIVE SUMMARY

Overall, Age, Sex, and MaxHR show strong patterns related to Heart Disease risk. Age is a clear risk factor, with disease risk increasing after 50. Female patients demonstrate higher prevalence across all age groups. MaxHR is a powerful physiological indicator, consistently lower among Heart Disease patients. These features are strong predictors and should be prioritized in further analysis and predictive modeling.

Clinical Symptoms Pattern



917
Total Reords



4
Chest Pain Type

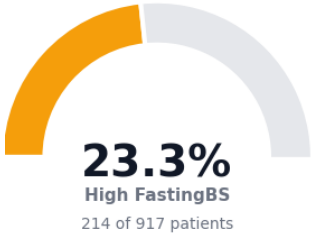


3
Exercise angina states

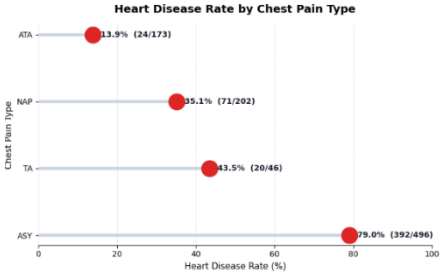


3
ST Slope Categories

GLUCOSE & CHEST PAIN OVERVIEW

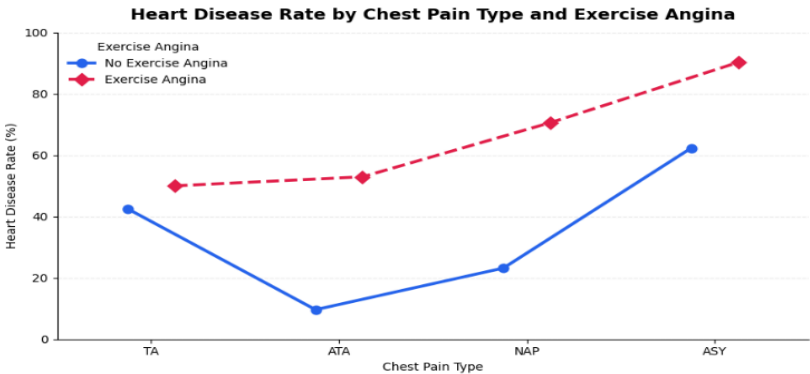


Elevated blood sugar appears in a meaningful portion of the dataset.



ASY shows the highest Heart Disease rate, indicating asymptomatic chest pain is strongly associated with higher risk.

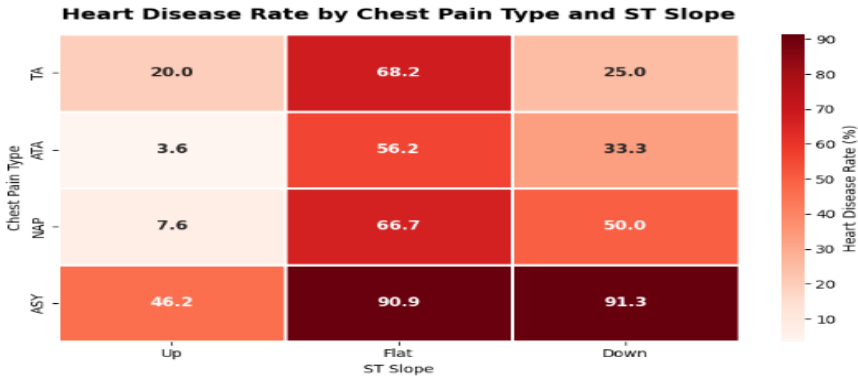
SYMPTOM INTERCATION



Patients with Exercise Angina consistently show higher Heart Disease rates across chest pain types

ECG SYMPTOM PATTERN

Flat and Down ST Slope patterns show higher Heart Disease rates, especially among ASY patients



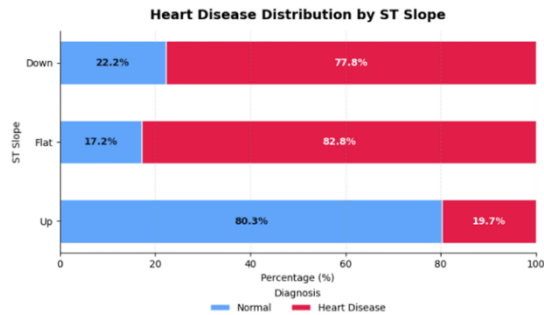
EXECUTIVE SUMMARY



Heart Disease risk becomes more visible when multiple symptoms appear together, especially ASY chest pain, Exercise Angina, and abnormal ST Slope patterns. These findings suggest that symptom combinations are more informative than individual indicators and should be prioritized in clinical assessment and predictive modeling

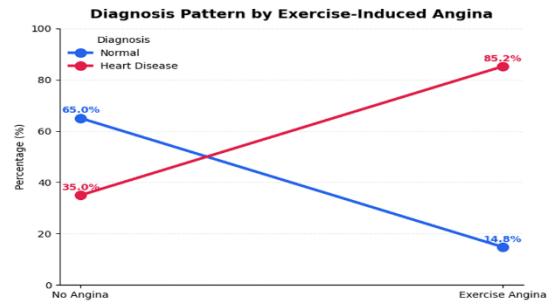
ECG, Exercise Response & Risk Summary

ECG BASELINE



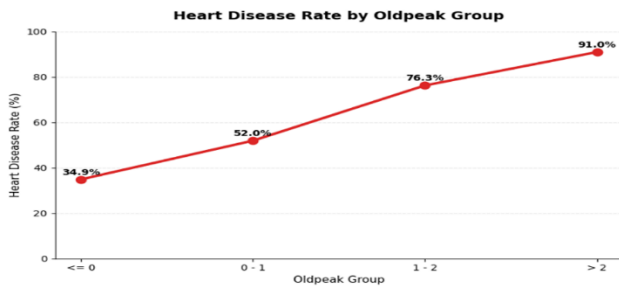
Flat and Down ST Slope show much higher Heart Disease proportions, making ST Slope one of the strongest clinical risk

EXERCISE RESPONSE

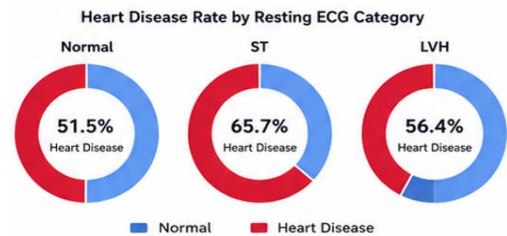


Patients with Exercise Angina are far more likely to have Heart Disease, showing a clear link between exercise-related chest pain and higher risk

EXECUTIVE SUMMARY



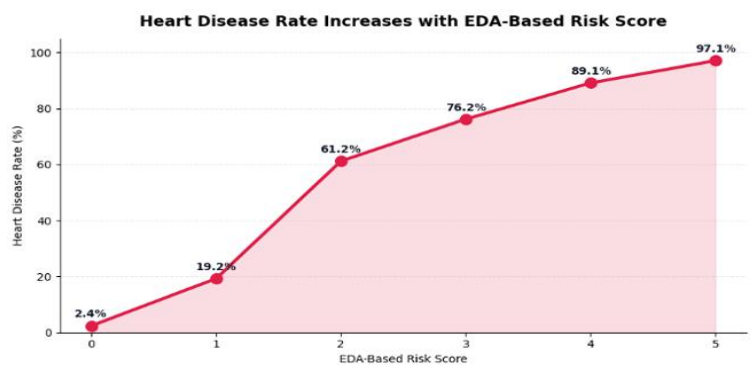
Heart Disease risk rises steadily as Oldpeak increases, indicating stronger ECG depression is linked to higher risk



Resting ECG categories show moderate differences, with ST and LVH having slightly higher Heart Disease rates than Normal

ECG SEVERITY & RESTING PATTERN

Heart Disease rate increases sharply as risk score rises, showing that combined EDA patterns can separate low-risk and high-risk patients well



EXECUTIVE SUMMARY



ECG and exercise-response features provide the clearest risk pattern: abnormal ST Slope, higher Oldpeak, and Exercise Angina consistently indicate higher Heart Disease risk. The combined EDA-based risk score clearly separates low-risk and high-risk patients, enabling stronger predictive decisions